

Hohmann Transfer Derivation

Part (a) calculates velocity changes required

Part (b) calculates time required

Part (c) calculates angular separation at start

The energy of a small body orbiting a much larger body is:

$$E = -\frac{GMm}{2a} = \frac{mv^2}{2} - \frac{GMm}{r}$$

where r is the distance from the primary focus and $\mu = GM$.
Solving for velocity, we get:

$$v^2 = \mu \cdot \left(\frac{2}{r} - \frac{1}{a} \right)$$

(a)

At the perigee:

$$v_{old} = \sqrt{\mu \cdot \left(\frac{2}{R} - \frac{1}{R} \right)} = \sqrt{\mu \cdot \frac{1}{R}}$$

and

$$v_{new} = \sqrt{\mu \cdot \left(\frac{2}{R} - \frac{2}{R+R'} \right)} = \sqrt{\mu \cdot \frac{2R'}{R(R+R')}}}$$

$$\therefore \Delta v = \sqrt{\frac{\mu}{R}} \left(\sqrt{\frac{2R'}{R+R'}} - 1 \right)$$

Similarly at the apogee:

$$\Delta v' = \sqrt{\frac{\mu}{R'}} \left(1 - \sqrt{\frac{2R}{R+R'}} \right)$$

(b)

From Kepler's Third Law we know that $T^2 = \frac{4\pi^2 a^3}{\mu}$

So the time taken is half the orbital time for the transfer ellipse:

$$T = \pi \cdot \sqrt{\frac{(R+R')^3}{8\mu}}$$

Let's calculate an example of Hohmann Transfer from Earth to Jupiter

$\therefore 1 \text{ Jupiter Year} = 12 \text{ Earth years}$

$$\Rightarrow R' = (144)^{1/3} R$$

So time taken will be:

$$T = \frac{1}{2} \text{ year} \times \sqrt{\frac{(1 + (144)^{1/3})^3}{2^3}}$$

$$\boxed{\Rightarrow T \approx 2.7565 \text{ years}}$$

(c)

Angular Velocity:

$$\omega = \sqrt{\frac{\mu}{(R')^3}}$$

So angular separation(α):

$$\boxed{\alpha = \pi - \omega T = \pi \cdot \left(1 - \frac{1}{2\sqrt{2}} \sqrt{\left(\frac{R}{R'} + 1\right)^3}\right)}$$

Alternatively if T_{orbit} is the time for one revolution of the target body then:

$$\alpha = \pi - \frac{T}{T_{orbit}} \cdot 2\pi$$

Let's use the same example again:

For Jupiter $T_{orbit} = 12$ years

$$\therefore \alpha = \pi - 2\pi \cdot \frac{2.7565}{12} \approx 1.6983 \text{ rad}$$

So the angular separation is $1.6983 \text{ rad} = 97.305^\circ$